Clinton Enwerem

enwerem@umd.edu | 301-416-9430 | clintonenwerem.com | GitHub: @coenwerem | LinkedIn | Scholar

Summary

Robotics Research & Software Engineer, Ph.D. candidate (ECE, University of Maryland, **degree expected Fall 2026**). Specialties in dexterous manipulation, risk-aware and safety-critical control, robot learning, motion planning, and ROS 2 software for physical multifingered systems.

Skills

Robotics: ROS/ROS2, ros2_control, MoveIt2, MuJoCo, Drake, Isaac Sim, Isaac Lab, Pinocchio, SocketCAN; **Motion Planning:** RRT*, Informed RRT*, Sampling-Based Planning, FCL Collision Checking; **Control & Safe Learning:** Control Barrier Functions, Safety Filters, Safe & Constrained Reinforcement Learning, Model Predictive Control (MPC), QP-based Control, Belief-Space Planning, Sim-to-Real Transfer; **Robot Learning & Generative Models:** Imitation Learning & Behavior Cloning, ACT (Action Chunking Transformers), DAgger, Diffusion Policies (DP3), Flow-Matching Generative Models, Reinforcement Learning (PPO); **ML / Perception:** PyTorch, TensorFlow, OpenCV; **Programming:** Python, C++, MATLAB, Bash; **Optimization:** CasADi, CVXPY, qpsolvers, IPOPT, Gurobi; **Dev Tools:** git, Docker; **Robots & EoATs:** Fairino FR3, xArm7, Unitree G1 (simulation), LEAP Hand v1, RealHand L6.

Work Experience

Graduate Research Assistant — Aug. 2021 – Present
Institute for Systems Research (ISR), University of Maryland — College Park, MD

- Build Python/C++ manipulation, planning, and control software for Fairino FR3 and xArm7 arms with LEAP Hand v1 and RealHand L6 end effectors.
- Build synchronized tactile, proprioceptive, and visual data pipelines and train behavior-cloning, recurrent, diffusion-policy, and reinforcement-learning baselines for contact-rich manipulation.

Robotics Trainee — Jan. 2020 – Feb. 2021
Robotics & Artificial Intelligence Nigeria — Ibadan, Nigeria

- Built ROS-based visual-SLAM, state-estimation, and flight-control software for a low-cost UAV on embedded hardware.

Projects

ParcelStow, Task-Rate Robustness for Dexterous Imitation (Paper, Code, Data) | *Isaac Lab, ACT, DAgger, Diffusion Policy* — 2026

- Built an Isaac Lab grasp-reorient-insert benchmark on a fixed-base Unitree G1 with a RealHand L6, collected **297** expert demonstrations, and trained DAgger, Diffusion Policy, and ACT baselines. ACT reached **100%** nominal success and fell to **53%** at 2× task rate against **84%** for the expert, a **31**-point gap inside the training-rate distribution.

`EquiDexFlow`, SE(3)-Equivariant Grasp Generation (Project Page, Code) | *Python, PyTorch* — 2026

- Trained an SE(3)-equivariant flow-matching model on **8,100** certified grasps across 81 objects, recording zero friction-cone violations and a **0.46 Nm** wrench residual, and retargeted generated grasps to a physical LEAP Hand by IK alone.

`realhand_l6_ros2`, Dexterous-Hand ROS 2 Stack (Code, DOI) | *C++, ROS2 Jazzy, ros2_control, SocketCAN, MoveIt2* — Aug. 2026

- Built a `hardware_interface::SystemInterface` SocketCAN driver, tactile contact-gated controller, robot description, and MoveIt 2 configuration, with mock and virtual-CAN CI. Validated CAN transport, hardware activation, joint-state feedback, tactile acquisition, contact detection, and contact-gated control on a physical right L6.

`kinematic_planner_ros2`, Sampling-Based Motion Planning (Code) | *ROS2, RRT*, Informed RRT*, FCL, Python* — 2026

- Implemented RRT* and Informed RRT* from first principles for N-DOF URDF manipulators with direct FCL primitive and triangle-mesh collision checks, a ROS-independent planner core, and benchmark CI. Benchmarked against MoveIt 2's OMPL, CHOMP, STOMP, and Pilz pipelines on a shared xArm7 query with per-waypoint revalidation.

Selected Publications

- **C. Enwerem**, J. S. Baras, C. Belta. "Grasp Execution Without a Planner: Configuration-Space Grasp Distance Fields with Certified Safety & Guaranteed Quality." *Preprint*, 2026. arXiv:2608.00600, project page.
- **C. Enwerem**, J. S. Baras, C. Belta. "Does Imitation Learning Preserve Temporal Robustness in Dexterous Manipulation? An Expert-Learner Comparison Across Task Execution Speeds." *Preprint*, 2026. arXiv:2609.01453, code.
- Also *to appear 2026* at IROS and CDC, plus 5 peer-reviewed first-author papers (CoDIT'23, L-CSS'24, ECC'24, CDC'24, CDC'25).

Education

Ph.D., Electrical & Computer Engineering, University of Maryland, College Park, MD. Expected Fall 2026. — Aug. 2021 – Present
B.Eng., Electrical Engineering, University of Nigeria, Nsukka. Highest Honors. GPA 3.84/4.0. — Aug. 2018